Multi-Disease Detection Report

CNN + LLM contextual reasoning • generated 2026-08-24 22:52

Patient context

Age	67
Sex	Male
Symptoms	Progressive memory loss, confusion, difficulty finding words
History	Symptoms gradually worsening over 2 years, family history of dementia

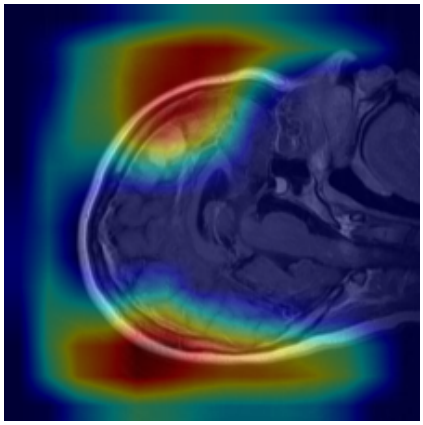
Model prediction

brain_meningioma **Malignant**

Brain Tumor · Neurological / Brain · Brain MRI

#	Condition	Confidence
1	brain_meningioma	85.9%
2	brain_pituitary	8.2%
3	brain_glioma	4.0%
4	oral_squamous_cell_carcinoma	0.6%
5	oral_normal	0.3%

Grad-CAM (model attention)



LLM contextual interpretation

Model suggests Brain Tumor (brain_meningioma) at 85.9% confidence.

Model finding. The CNN's top class is 'brain_meningioma' (Malignant, Neurological / Brain) on Brain MRI, at 85.9%.

Confidence. High model confidence (85.9%) with a clear 77.7 pt margin over the runner-up.

Differential

- **brain_meningioma** — ranked #1 at 85.9%
- **brain_pituitary** — ranked #2 at 8.2%
- **brain_glioma** — ranked #3 at 4.0%

Grad-CAM correlation. Model attention was a moderately sized region in the central portion of the image (~21% high-salience area). Verify this region is anatomically plausible for Brain MRI.

Biomedical context. [KB] Meningioma — usually slower-growing tumor of the meninges. Group: Brain Tumor; organ: Neurological / Brain; modality: Brain MRI; category: Malignant (severity: moderate). Primary brain tumors include

gliomas, meningiomas, and pituitary adenomas, differing in location, growth pattern, and MRI appearance; imaging plus histology establish the diagnosis. [BioGPT fallback — set ENABLE_BIOGPT=true to run the real model] MRI with contrast characterizes the lesion; definitive typing requires histopathology/molecular markers, often after surgical sampling. [Med-PaLM 2: PLACEHOLDER — no public API; see medpalm_adapter.py to wire MedLM]

Patient considerations. Provided context (age: 67; sex: Male; symptoms: Progressive memory loss, confusion, difficulty finding words; history: Symptoms gradually worsening over 2 years, family history of dementia) should be correlated with the finding.

Recommended next steps

- Correlate with the full clinical picture and prior imaging.
- Obtain gold-standard confirmation appropriate to the modality (e.g., biopsy/histology or specialist review).
- Escalate to the relevant specialist if the category is malignant or high-severity.

Uncertainty & limitations

- Interpretation generated WITHOUT the Claude reasoning model (offline fallback).
- Model trained on specific public datasets; performance may not transfer to other scanners/populations.
- A confident softmax score is not calibrated probability of disease.

■ Research decision-support only — not a medical diagnosis. All findings require confirmation by a licensed clinician.